Regional Locking Inference: Geometry Chain and Density Predictions

域内锁定推断：几何链与密度预言

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Abstract.
Starting from the single axiom A1, the geometry chain is derived without external input. The minimum stable closure is a three-node structure; its incircle gives $a = \pi/(3\sqrt{3})$. Four density parameters follow from integer pairings of u_0^2 and w_0^2, all within 1σ of Planck 2018. The three-layer lock yields biological and symbolic frequencies matching EEG observations to 0.06% and 0.21%.

摘要.
从唯一公理 A1 出发，几何链不借外部输入。最小稳定闭合为三节点结构，其内切圆给出 $a = \pi/(3\sqrt{3})$。四条密度从 u_0^2 与 w_0^2 的整数配对推出，全落 Planck 2018 的 1σ 内。三层锁给出生物层与符号层频率，对上脑电观测 0.06% 与 0.21%。

I. Axiom and Constraint Chain / 公理与约束链

A1. Measurement is the process of establishing a relational lock between two intrinsic locking structures. No space, time, particle, or force is presupposed.

A1. 测量是两个内禀锁定结构之间建立关系锁定的过程。不预设空间、时间、粒子、力。

读法

Constraint 0: before a relational lock is established, the two ends are indistinguishable.

约束零：建立关系锁定之前，两端不可区分。

读法

Constraint 1: closure must be stable.

约束一：闭合必须稳定。

读法

Constraint 2: stability requires redundancy.

约束二：稳定要求冗余。

推导

Constraint 3: redundancy is minimal.

约束三：冗余取最小。

Chain: A1 → Constraint 0 → shared resource pool → Constraint 1 → Constraint 2 → Constraint 3 → three-node closure.

链条：A1 → 约束零 → 资源池共享 → 约束一 → 约束二 → 约束三 → 三节点闭合。

II. Geometry Chain / 几何链

Three-node closure → equilateral triangle → incircle.

三节点闭合 → 等边三角形 → 内切圆。

$$a = \frac{\pi}{3\sqrt{3}} = 0.6045997881$$

$$u_0 = 1 - a = 0.3954002119, \quad u_0^2 = 0.1563413274$$

$$w_0^2 = 1 - u_0^2 = 0.8436586726, \quad \beta = 4u_0^2 = 0.6253653097$$

$$\theta_t = 2\arccos(1 - 4u_0^2) = 2.37383950$$

选择

Choice: undirected edge for the non-returning pair.

选择：二不成回路取无向边。

读法

Reading: "on a plane" and "two-dimensional common region".

读法：“在平面上”与“共同区二维性”。

III. Four Densities / 四条密度

Four pairings: 3D×3D, 3D×4D, 4D×3D, 4D×4D. Coefficients $c_1 = 2,\ c_2 = 1,\ c_3 = -1,\ c_4 = 2$.

四种配对：3D×3D、3D×4D、4D×3D、4D×4D。系数 $c_1 = 2,\ c_2 = 1,\ c_3 = -1,\ c_4 = 2$。

$$\Omega_b = 2u_0^4 = 0.04888522$$

$$\Omega_{DM} = 2u_0^2 w_0^2 = 0.26379743$$

$$\Omega_m = 2u_0^2 = 0.31268265$$

$$\Omega_\Lambda = w_0^2 - u_0^2 = 0.68731735$$

Additive closure: $\Omega_b + \Omega_{DM} = \Omega_m$, $\Omega_m + \Omega_\Lambda = 1$. Uniqueness of coefficients is conditional on the four-pairing reading.

加法闭合：$\Omega_b + \Omega_{DM} = \Omega_m$，$\Omega_m + \Omega_\Lambda = 1$。系数唯一性条件于四种配对读法。

Core prediction: $\Omega_b/\Omega_m = u_0^2 = 0.1563413274$.

核心预言： $\Omega_b/\Omega_m = u_0^2 = 0.1563413274$。

IV. Planck 2018 Collision / Planck 2018 对撞

Quantity / 量	RLI	Planck 2018	Deviation / 偏差
Ω_b	0.04888522	0.0493 ± 0.0006	0.68σ
Ω_{DM}	0.26379743	0.2650 ± 0.0073	0.16σ
Ω_m	0.31268265	0.3153 ± 0.0073	0.36σ
Ω_Λ	0.68731735	0.6847 ± 0.0073	0.36σ

Ω_b/Ω_m Path / 路径	Observed / 观测	Deviation / 偏差
Planck	0.1573 ± 0.006	0.16σ
BBN + Planck	0.1561	0.04σ
BOSS EFT	0.154 ± 0.022	0.10σ
BOSS pre-recon / 重建前	0.153 ± 0.029	0.11σ
BOSS post-recon / 重建后	0.173 ± 0.027	0.62σ

V. Layer Lock Frequencies / 层锁频率

Three layers with vertical inheritance: $\phi_{\text{phys}} = 0$, $\phi_{\text{bio}} = \delta = 2/9$, $\phi_{\text{sym}} = 2\delta = 4/9$.

三层垂直继承：$\phi_{\text{phys}} = 0$，$\phi_{\text{bio}} = \delta = 2/9$，$\phi_{\text{sym}} = 2\delta = 4/9$。

$$\Gamma_{\text{bio}} = 2 \cdot 3^3 \cdot \delta = 54\delta = 12 \text{ Hz}$$

$$\Gamma_{\text{sym}} = 2^2 \cdot 5 \cdot \delta = 20\delta = 4.444 \text{ Hz}$$

$$\gamma_{bg,i} = \Gamma_i \left(1 - \cos(\theta_t/2 + \phi_i)\right)$$

At $\theta_t = 2.37384$, $\theta_t/2 = 1.18692$.

取 $\theta_t = 2.37384$，$\theta_t/2 = 1.18692$。

Biological layer: $1.18692 + 0.22222 = 1.40914$, $\cos = 0.16117$, $1 - \cos = 0.83883$.

$\gamma_{\text{bio}} = 12 \times 0.83883 = 10.066$ Hz. Observed 10.06 Hz, deviation 0.06%.

生物层：$1.18692 + 0.22222 = 1.40914$，$\cos = 0.16117$，$1 - \cos = 0.83883$。

$\gamma_{\text{bio}} = 12 \times 0.83883 = 10.066$ Hz。观测 10.06 Hz，差 0.06%。

Symbolic layer: $1.18692 + 0.44444 = 1.63136$, $\cos = -0.05977$, $1 - \cos = 1.05977$.

$\gamma_{\text{sym}} = 4.44444 \times 1.05977 = 4.7101$ Hz. Observed 4.70 Hz, deviation 0.21%.

符号层：$1.18692 + 0.44444 = 1.63136$，$\cos = -0.05977$，$1 - \cos = 1.05977$。

$\gamma_{\text{sym}} = 4.44444 \times 1.05977 = 4.7101$ Hz。观测 4.70 Hz，差 0.21%。

VI. Hierarchy / 层级

推导

A1, three nodes, a, u_0^2, four densities, Ω_b/Ω_m, Γ_{bio}, Γ_{sym}, $\gamma_{bg,i}$.

A1、三节点、a、u_0^2、四条密度、Ω_b/Ω_m、Γ_{bio}、Γ_{sym}、$\gamma_{bg,i}$。

读法

Constraints 0–2, "on a plane", "two-dimensional common region".

约束零一二、“在平面上”、“共同区二维性”。

选择

Undirected edge choice; decomposition 54δ and 20δ.

二不成回路取无向边；54δ 与 20δ 的分解。

候选/接口（另文）

$\alpha = \sqrt{3}/236$ (rooted in 118 via Madelung), $1/6$, $\Gamma_{AA} = 20/27$, glueball 7.5, vacuum energy density, lepton mass ratios, M_R. Not in this paper.

$\alpha = \sqrt{3}/236$（根在 118，依赖 Madelung）、$1/6$、$\Gamma_{AA} = 20/27$、胶球 7.5、真空能密度、轻子质量比、M_R。这些不在本文。

边界

$\gamma = 0$, before the break, mechanism of the break, value judgments, Qualia. Not answered.

$\gamma = 0$、破之前、破的机制、价值判断、Qualia。不回答。